

05.1 Describe the structure of the human immunodeficiency virus (HIV).

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[4 marks]

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Question 5 continues on the next page

Some people infected with HIV do not develop AIDS. These people are called HIV controllers.

Scientists measured the number of HIV particles (the viral load) and the number of one type of T helper cell (CD4 cells) in the blood of a group of HIV controllers and also in a group of HIV positive patients who had symptoms of AIDS.

The median values and the range of their results are shown in **Table 3**.

Table 3

HIV status of people	Median viral load / virus particles per cm^3 of blood (range)	Median number of CD4 cells per mm^3 of blood (range)
HIV controllers	212 (<50 to 609)	693 (529 to 887)
HIV positive people with AIDS symptoms	66 274 (30 206 to 306 163)	248 (107 to 365)

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A test sample of 500 mm^3 of blood is taken from an HIV controller to determine the viral load.

Tick (✓) **one** box that shows the number of virus particles that would be present in a test sample of blood taken from an HIV controller with the median viral load.

[1 mark]

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106

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